

AI Document Intelligence Assistant (RAG)

1. Problem Statement

Title	Description
Problem Statement	Organizations and students often spend significant time searching through lengthy documents such as PDFs, reports, manuals, and policies. Traditional keyword search cannot understand context, making information retrieval slow and inefficient.

2. Problem Definition

Aspect	Description
Existing Problem	Manual document reading is time-consuming, and general LLMs may generate inaccurate answers without access to private documents.
Proposed Solution	Build an AI-powered Document Intelligence Assistant using Retrieval-Augmented Generation (RAG) that retrieves relevant document sections before generating answers.
Outcome	Enables users to ask natural language questions and receive accurate, context-aware responses from their uploaded documents.

3. Tech Stack

Category	Technology	Purpose
Frontend	Streamlit	User interface for document upload and chat
Backend	FastAPI	API for handling requests and RAG pipeline
Programming Language	Python	Core application development
LLM	Google Gemini API	Generates context-aware answers
AI Framework	LangChain	Orchestrates the RAG workflow
Embedding Model	Sentence Transformers	Converts text into vector embeddings

Category	Technology	Purpose
Vector Database	FAISS	Stores and retrieves document embeddings efficiently
Deployment	Streamlit Cloud + Render	Hosts frontend and backend
Version Control	Git & GitHub	Source code management and collaboration

4. Key Features

Feature	Description
Document Upload	Upload PDF documents for analysis
Semantic Search	Finds relevant content based on meaning rather than keywords
AI Question Answering	Answers questions using document context
RAG Pipeline	Combines retrieval with LLM generation for higher accuracy
Interactive Chat	Conversational interface for querying documents